

Security Log Normalization & Risk Exposure Platform

Abstract

This report documents the design, implementation, and outcomes of the Security Log Normalization & Risk Exposure Platform developed during the Cybersecurity Internship at CCTS Global. The project addresses challenges in heterogeneous security log analysis by standardizing log data, computing normalized risk scores, and visualizing exposure metrics through an interactive dashboard. The system enables consistent risk assessment, improved analyst efficiency, and informed decision-making.

1. Introduction

Modern organizations generate large volumes of security logs from diverse sources such as Windows Event Logs (EVTX), Linux system logs, intrusion detection systems (IDS), and vulnerability scanners. These logs vary in structure, severity scales, and context, making unified analysis difficult. This project was undertaken to normalize disparate log formats into a common schema and to quantify risk exposure in a consistent, explainable manner.

2. Objectives

- To collect and ingest heterogeneous security log sources.
- To normalize log fields into a unified data model.
- To calculate a normalized risk score per event.
- To present insights via a web-based dashboard.
- To support prioritization of security incidents and remediation.

3. Background and Rationale

Security Operations Centres (SOCs) rely on timely and accurate log analysis to detect threats. Without normalization, comparing severity across tools is unreliable and slows response. Normalization enables apples-to-apples comparison, correlation across sources, and scalable analytics. The risk exposure platform was built to demonstrate these principles in a practical system.

4. System Architecture

The platform follows a client–server architecture:

- **Backend:** Python-based service for log parsing, normalization, and risk scoring
- **Frontend:** React application styled with Tailwind CSS for visualization
- **Data Flow:** Log upload → Parsing → Normalization → Risk Scoring → Dashboard Rendering

5. Data Sources and Ingestion

Supported inputs include:

- Windows EVTX security logs
- Linux authentication and system logs
- IDS alerts
- Vulnerability scan outputs (CSV)

Uploaded files are processed server-side and transformed into a standardized dataset.

6. Normalization Methodology

Each log entry is mapped to a unified schema containing:

- Asset identifier and type
- Vulnerability or event identifier
- Severity
- Likelihood
- Business impact
- Owner
- Timestamp of last detection

Normalization ensures consistent ranges and data types, enabling accurate aggregation and analysis.

7. Risk Scoring Model

A normalized risk score is computed using weighted factors such as severity, likelihood, and business impact. Scores are scaled to a common range to allow prioritization. This model provides transparency and can be adjusted based on organizational risk appetite.

8. Frontend Dashboard

The dashboard provides:

- File upload interface for log ingestion
- Summary metrics (total events, high-risk events, assets, users)
- Tabular view of normalized events
- Visual cues to highlight high-risk items

9. Results and Outcomes

The platform successfully normalized thousands of log entries into a consistent format and produced uniform risk scores. Analysts can quickly identify high-risk events and affected assets, demonstrating improved efficiency compared to manual review.

10. Proof-of-Concept Scope and Constraints

This platform was developed as a proof of concept to validate the feasibility of security log normalization and risk exposure analysis. To maintain clarity and focus, the scope was intentionally limited to core functionalities:

- The risk scoring model uses predefined, static weights to demonstrate normalization and prioritization logic.
- Automated correlation between related events was limited to individual event analysis.
- Visualizations focus on tabular representations to emphasize data accuracy over advanced analytics.

11. Production-Scale Considerations and Future Work

If extended beyond a proof of concept, the platform can be enhanced to support enterprise-level security operations, including:

- Dynamic or machine learning–based risk scoring models that adapt to evolving threats.
- Advanced event correlation and automated alerting across multiple log sources.
- Role-based access control for SOC and management users.
- Expanded dashboards with graphical analytics and reporting capabilities.

12. Conclusion

The Security Log Normalization & Risk Exposure Platform successfully demonstrates the feasibility of a proof-of-concept system for security log normalization and risk assessment. The platform validates that heterogeneous logs from Windows, Linux, IDS, and vulnerability scanners can be standardized, scored for risk, and visualized in a clear dashboard format.

As a proof-of-concept, the project focuses on core functionalities, providing a working model of normalization, risk scoring, and dashboard visualization without full production-scale features. The outcomes confirm that such a platform can improve analyst efficiency and enable clear prioritization of high-risk events.

This PoC serves as a foundation for future enhancements, including dynamic risk scoring, event correlation, automated alerting, and advanced visualizations, illustrating the potential for scaling to enterprise-level security operations.

13. Tools and Technologies

- Python (backend processing)
- Pandas (data manipulation)
- React (frontend framework)
- Tailwind CSS (styling)
- REST APIs (backend–frontend integration)